Explicit Gaussian constants for unconditional multiplier rescaling

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Status. Partial result likely valid: an explicit quantitative strengthening of a later literature answer. The original qualitative conjecture is already resolved by Tselishchev [2].

Source conjecture and literature boundary

Balazs, Freeman, Popescu, and Speckbacher formulate the following finite conjecture [1, Conjecture 1.2, p. 3]. If finite sequences $(x_j)_{j=1}^N, (f_j)_{j=1}^N$ in a finite-dimensional complex Hilbert space H, symbols $(m_j)_{j=1}^N$, and $C > 0$ satisfy

$$\left\| \sum_{j=1}^{N} \varepsilon_j m_j \langle u, f_j \rangle x_j \right\| \le C\|u\| \qquad (u \in H,\ |\varepsilon_j| = 1), \tag{1}$$

does there exist a universal κ and factorizations $c_j\overline{d_j} = m_j$ such that both $(c_j x_j)$ and $(d_j f_j)$ have Bessel bound κC?

> been applied to solve many other problems in applied harmonic analysis and approximation theory [NOU][FS][LT][DKU]. In Section 2 we show that Conjecture 1.1 is equivalent to the following quantitative and finite-dimensional conjecture.
>
> **Conjecture 1.2.** *There exists a universal constant $\kappa > 0$ so that the following holds. Let $C > 0$ and let $M_{(m_j)_{j=1}^N,(x_j)_{j=1}^N,(f_j)_{j=1}^N}$ be a multiplier on a finite dimensional Hilbert space H such that*
>
> $$\Big\| \sum_{j=1}^{N} \varepsilon_j m_j \langle x, f_j \rangle x_j \Big\| \le C\|x\|, \qquad \textit{for all } x \in H \textit{ and } |\varepsilon_j| = 1. \tag{1.4}$$
>
> *Then there exists sequences of constants $(c_j)_{j=1}^N$ and $(d_j)_{j=1}^N$ such that $c_j\overline{d_j} = m_j$ for all $1 \le j \le N$ and both $(c_j x_j)_{j=1}^N$ and $(d_j f_j)_{j=1}^N$ are $C\kappa$-Bessel.*

Tselishchev subsequently proved this conjecture, in the equivalent nonzero-vector formulation, and thereby settled the corresponding infinite-dimensional rescaling problem [2, Theorems 2.1–2.3]. His Rademacher proof gives $\kappa = 2$. Remark 2.6(1) says that replacing the Rademacher variables by Gaussian variables gives a better constant, but it does not state that constant and leaves evaluation of the best constant beyond the paper's scope. The result below evaluates the Gaussian argument and combines it with a direct convex balancing step.

We use the convention that a sequence (z_j) is B-Bessel when

$$\sum_j |\langle u, z_j \rangle|^2 \le B\|u\|^2 \qquad (u \in H).$$

Proof intuition

After absorbing the symbols m_j into the two vector sequences, the phase estimate (1) says that every scalar matrix coefficient has an ℓ_1 bound. Test this bound on two independent Gaussian vectors with arbitrary covariance density matrices ρ and σ. The exact first absolute moment of a standard circular complex Gaussian is $\sqrt{\pi}/2$; hence the resulting Hellinger sum is bounded by $(4/\pi)C$.

The remaining problem is to select one scalar weight per rank-one term so that two global operator norms are simultaneously small. Write the weights as $e^{s_j/2}$ and minimize the sum of those two norms over $s \in \mathbb{R}^N$. At a minimizer, the subgradient equation supplies top-eigenspace density matrices ρ, σ for which the two weighted contributions agree for every j. Thus each of the two operator norms becomes exactly the Hellinger sum controlled by Gaussian averaging.

The quantitative result

Theorem 1 (Explicit complex Gaussian constant)**.** *Assume* (1)*. If every term with $m_j \neq 0$ has $x_j \neq 0$ and $f_j \neq 0$, then there are scalars c_j, d_j satisfying $c_j\overline{d_j} = m_j$ such that both $(c_j x_j)_{j=1}^N$ and $(d_j f_j)_{j=1}^N$ are $(4C/\pi)$-Bessel.*

Without the nondegeneracy assumption, the same conclusion holds with $(4/\pi + \eta)C$ in place of $4C/\pi$ for every $\eta > 0$. In particular, Conjecture 1.2 holds for arbitrary finite data with the explicit universal constant $\kappa = 3/2$.

The proof is split into the mixed Gaussian estimate and the balancing lemma.

Lemma 1 (Gaussian Hellinger estimate)**.** *Let $(X_j), (Y_j)$ be nonzero vectors in a finite-dimensional complex Hilbert space such that*

$$\left\| \sum_{j=1}^N \varepsilon_j \langle u, Y_j \rangle X_j \right\| \leq C\|u\| \qquad (u \in H,\ |\varepsilon_j| = 1). \tag{2}$$

Put $P_j = X_j \otimes X_j$ and $Q_j = Y_j \otimes Y_j$. For all density matrices $\rho, \sigma \in \mathcal{D}(H) := \{R \geq 0 : \operatorname{Tr} R = 1\}$,

$$\sum_{j=1}^N \sqrt{\operatorname{Tr}(\rho P_j)\operatorname{Tr}(\sigma Q_j)} \leq \frac{4}{\pi} C. \tag{3}$$

Proof. Taking a scalar matrix coefficient in (2) and choosing the phases to align its summands gives, for all $u, v \in H$,

$$\sum_{j=1}^N |\langle u, Y_j \rangle|\,|\langle X_j, v \rangle| \leq C\|u\|\|v\|. \tag{4}$$

Let G_σ, G_ρ be independent centered circular complex Gaussian vectors with covariance matrices σ, ρ, respectively. If γ is a scalar circular complex Gaussian normalized by $\mathbb{E}|\gamma|^2 = 1$, then polar integration gives

$$\mathbb{E}|\gamma| = \frac{\sqrt{\pi}}{2}. \tag{5}$$

Consequently,

$$\mathbb{E}|\langle G_\sigma, Y_j \rangle| = \frac{\sqrt{\pi}}{2}\sqrt{\operatorname{Tr}(\sigma Q_j)}, \qquad \mathbb{E}|\langle X_j, G_\rho \rangle| = \frac{\sqrt{\pi}}{2}\sqrt{\operatorname{Tr}(\rho P_j)}.$$

Apply (4) to (G_σ, G_ρ) and take expectations. Independence and the two identities above make the left side $\frac{\pi}{4}$ times the left side of (3). On the other hand, Cauchy–Schwarz gives

$$\mathbb{E}\|G_\sigma\| \le (\mathbb{E}\|G_\sigma\|^2)^{1/2} = (\operatorname{Tr}\sigma)^{1/2} = 1,$$

and likewise for G_ρ. This proves (3). □

Lemma 2 (Convex balancing)**.** *Suppose P_j, Q_j are nonzero positive rank-one operators and, for some $L > 0$,*

$$\sum_j \sqrt{\operatorname{Tr}(\rho P_j)\operatorname{Tr}(\sigma Q_j)} \le L \qquad (\rho, \sigma \in \mathcal{D}(H)). \tag{6}$$

Then there are positive numbers α_j such that

$$\Big\|\sum_j \alpha_j^2 P_j\Big\| \le L, \qquad \Big\|\sum_j \alpha_j^{-2} Q_j\Big\| \le L. \tag{7}$$

Proof. For $s = (s_1, \dots, s_N) \in \mathbb{R}^N$ define

$$A(s) = \lambda_{\max}\left(\sum_j e^{s_j} P_j\right), \qquad B(s) = \lambda_{\max}\left(\sum_j e^{-s_j} Q_j\right).$$

Both functions are convex. Since every P_j, Q_j is nonzero, $A(s)+B(s)$ is coercive: if some $s_j \to +\infty$, then $A(s) \ge e^{s_j}\|P_j\| \to \infty$, while if some $s_j \to -\infty$, then $B(s) \ge e^{-s_j}\|Q_j\| \to \infty$. Thus $A+B$ has a minimizer s^*.

The variational formula for the top eigenvalue is

$$A(s) = \max_{\rho\in\mathcal{D}(H)} \sum_j e^{s_j}\operatorname{Tr}(\rho P_j), \tag{8}$$

and analogously for B. The subdifferential of A at s consists of the convex hull of the vectors

$$\big(e^{s_j}\operatorname{Tr}(\rho P_j)\big)_{j=1}^N$$

coming from maximizers ρ in (8). A convex combination of maximizing density matrices is again a maximizing density matrix, so the convex hull is already represented by a single maximizing ρ. The corresponding statement for B gives subgradients

$$\big(-e^{-s_j}\operatorname{Tr}(\sigma Q_j)\big)_{j=1}^N$$

with σ maximizing the formula for B.

The optimality condition $0 \in \partial A(s^*) + \partial B(s^*)$ therefore provides maximizing density matrices ρ, σ such that, for every j,

$$e^{s_j^*}\operatorname{Tr}(\rho P_j) = e^{-s_j^*}\operatorname{Tr}(\sigma Q_j). \tag{9}$$

Put $a_j = \operatorname{Tr}(\rho P_j)$ and $b_j = \operatorname{Tr}(\sigma Q_j)$. Since ρ and σ are maximizing, (9) yields

$$A(s^*) = \sum_j e^{s_j^*} a_j = \sum_j \sqrt{a_j b_j} \le L,$$

and identically $B(s^*) = \sum_j \sqrt{a_j b_j} \le L$. Taking $\alpha_j = e^{s_j^*/2}$ proves (7). □

Proof of Theorem 1. For $m_j \neq 0$, write $m_j = |m_j|\omega_j$, $|\omega_j| = 1$, and set

$$X_j = \omega_j \sqrt{|m_j|}\, x_j, \qquad Y_j = \sqrt{|m_j|}\, f_j.$$

Then $\langle u, Y_j\rangle X_j = m_j \langle u, f_j\rangle x_j$, so (1) becomes (2). Terms with $m_j = 0$ can be discarded temporarily and later assigned $c_j = d_j = 0$.

Under the nondegeneracy assumption, Lemma 1 and Lemma 2, with $L = 4C/\pi$, give positive α_j for which $(\alpha_j X_j)$ and $(\alpha_j^{-1} Y_j)$ are both $(4C/\pi)$-Bessel. Define

$$c_j = \omega_j \sqrt{|m_j|}\, \alpha_j, \qquad d_j = \sqrt{|m_j|}\, \alpha_j^{-1}.$$

Then $c_j \overline{d_j} = m_j$, and the two scaled sequences are exactly $(\alpha_j X_j)$ and $(\alpha_j^{-1} Y_j)$.

For completeness, suppose now that $m_j \neq 0$ but one of x_j, f_j is zero. Such an index contributes nothing to the multiplier. If $x_j = 0$ and $f_j \neq 0$, choose d_j arbitrarily small and put $c_j = m_j/\overline{d_j}$; only the frame operator on the f side receives a nonzero contribution. If $f_j = 0$ and $x_j \neq 0$, do the symmetric construction. If both vectors are zero, choose any finite factorization of m_j. Since there are finitely many degenerate indices, their total positive contributions can be made to have norm at most ηC on each side. Adding them to the nondegenerate part proves the $(4/\pi + \eta)C$ bound. Since $4/\pi < 3/2$, take $\eta = 3/2 - 4/\pi$ to obtain the last assertion. $\square$

Corollary 1 (Explicit rank-one completely bounded constant)**.** *Let H be a finite-dimensional complex Hilbert space, and let $\Phi : \ell_\infty^N \to B(H)$ be linear with* $\operatorname{rank} \Phi(e_j) \leq 1$ *for every coordinate vector e_j. Then*

$$\|\Phi\|_{\mathrm{cb}} \leq \frac{4}{\pi} \|\Phi\|.$$

Proof. Write

$$\Phi(a)u = \sum_{k=1}^{N} a(k) \langle u, Y_k\rangle X_k$$

and put $C = \|\Phi\|$. As in (4), for all $u, v \in H$,

$$\sum_k |\langle u, Y_k\rangle|\, |\langle X_k, v\rangle| \leq C\|u\|\|v\|. \tag{10}$$

Fix m and vectors $u_1, \dots, u_m, v_1, \dots, v_m \in H$. Substitute $u = \sum_j \gamma_j u_j$ and $v = \sum_i \delta_i v_i$ in (10), where the γ_j and δ_i are two independent families of standard circular complex Gaussians. Taking expectations and using (5) gives

$$\sum_k \left(\sum_j |\langle u_j, Y_k\rangle|^2 \right)^{1/2} \left(\sum_i |\langle X_k, v_i\rangle|^2 \right)^{1/2} \leq \frac{4}{\pi} C \left(\sum_j \|u_j\|^2 \right)^{1/2} \left(\sum_i \|v_i\|^2 \right)^{1/2}. \tag{11}$$

An element of $M_m(\ell_\infty^N)$ is a tuple $(A_k)_{k=1}^N$ of $m \times m$ matrices, with norm $\max_k \|A_k\|$. In a matrix coefficient of $\Phi_{(m)}((A_k))$, the contribution belonging to k is the trace pairing of A_k with the rank-one matrix

$$B_k = \big(\langle u_j, Y_k\rangle \langle X_k, v_i\rangle \big)_{i,j=1}^m.$$

Its trace norm is

$$\|B_k\|_1 = \left(\sum_j |\langle u_j, Y_k\rangle|^2 \right)^{1/2} \left(\sum_i |\langle X_k, v_i\rangle|^2 \right)^{1/2}.$$

Therefore trace duality and (11) show that

$$\|\Phi_{(m)}((A_k))\| \leq \frac{4}{\pi} C \max_k \|A_k\|.$$

The estimate is independent of m, proving the assertion. □

Corollary 2 (Real variant)**.** *Over a real Hilbert space, the same proof gives the exact nondegenerate constant $\kappa = \pi/2$, and every $\kappa > \pi/2$ for arbitrary finite data.*

Proof. Use independent real standard Gaussians in Lemma 1. The identity $\mathbb{E}|g| = \sqrt{2/\pi}$ replaces (5), so the product of the two first moments is $2/\pi$ and the Hellinger constant becomes $\pi/2$. The balancing proof is unchanged. □

Verification notes and limitations

- The factors $4/\pi$ and $\pi/2$ come from exact first absolute Gaussian moments; no numerical approximation is used.
- The coercivity assertion is why nonzero X_j, Y_j are separated from irrelevant zero-vector terms. The latter are restored with arbitrarily small positive frame-operator contributions, giving every constant strictly larger than the displayed infimum.
- The subgradient step permits multiple top eigenvalues: convex combinations of top-eigenspace vector states are density matrices supported on the same top eigenspace, so no differentiability assumption is hidden.
- This packet does not determine the optimal universal constant. The one-term example only gives the elementary lower bound $\kappa \geq 1$.
- The broad conjecture is not new: it was solved in arXiv:2508.02802. The candidate new content is the evaluated constant and the direct convex-balancing proof. Corollary 1 evaluates the same Gaussian refinement for the later paper's equivalent completely bounded-map formulation.

Bounded novelty check

The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2212.00947 and the core terms "unconditional pairs", "frame multiplier", "rescaling", and "Bessel". The local 2000–present arXiv source corpus and forward-title searches identified arXiv:2508.02802 as the later solution. Its Theorem 2.2, proof, and Remark 2.6(1) were checked directly. Exact web searches on July 19, 2026 used combinations of "$4/\pi$", "$\pi/2$", "Gaussian", "unconditional Schauder frame", "rank-one completely bounded map", and both arXiv ids. No source stating either explicit constant for this rescaling problem was found; the decisive later paper only says that the Gaussian constant is better than 2.

Novelty confidence is therefore *moderate*: the explicit computation was not found in the checked literature, but it is a natural completion of a Gaussian refinement already signposted by Tselishchev. The recommended human review should focus on the convex subdifferential balancing lemma and on whether the constants have appeared elsewhere under the equivalent completely bounded-map formulation.

References

[1] P. Balazs, D. Freeman, R. Popescu, and M. Speckbacher, *Quantitative bounds for unconditional pairs of frames*, arXiv:2212.00947 (2022).

[2] A. Tselishchev, *Rescaling of unconditional Schauder frames in Hilbert spaces and completely bounded maps*, arXiv:2508.02802 (2025).